CIFAR-10, CIFAR-100, and STL-10 datasets using both S-Mixer and RP S-Mixer architectures with $S = C = 64$, along with four blocks. For optimization, we set the initial learning rate to be 0.1.

For CIFAR-10 and CIFAR-100, we trained the models for 200 epochs and evaluated them with different patch sizes ($P = 1, 2, 4, 8, 16, 32$). We performed three trials with different random seeds for each setting. On the STL-10 dataset, we resized the images to $90 \times 90 \times 3$ and trained the models for 400 epochs. We varied the patch size ($P = 1, 3, 6, 10, 18, 30$) and performed five trials with different random seeds for each setting.

D.7. RP can prevent the overfitting

In Figure S.6 , we explored several values of C fixed Ω, the normal model shows overfitting of more than twice the magnitude compared to the RP model, especially C is the largest one in the exploring range. In this case, S takes the smallest value among the explored values. This suggests that RP has a regularization effect beyond the token-mixing side and affects the channel-mixing side, particularly when C is large.

To mitigate overfitting, additional augmentation (auto-augmentation, based on (Cubuk et al., 2019)) was applied to the dataset, and the model was switched from S-Mixer to MLP-Mixer ($\gamma = 4$) due to the observed slower decrease in training loss for S-Mixer in Figure S.7 . RP S-Mixer outperformed S-Mixer in terms of test loss for $C = 173$, indicating that RP still provides overfitting prevention even with relatively strong data augmentations.

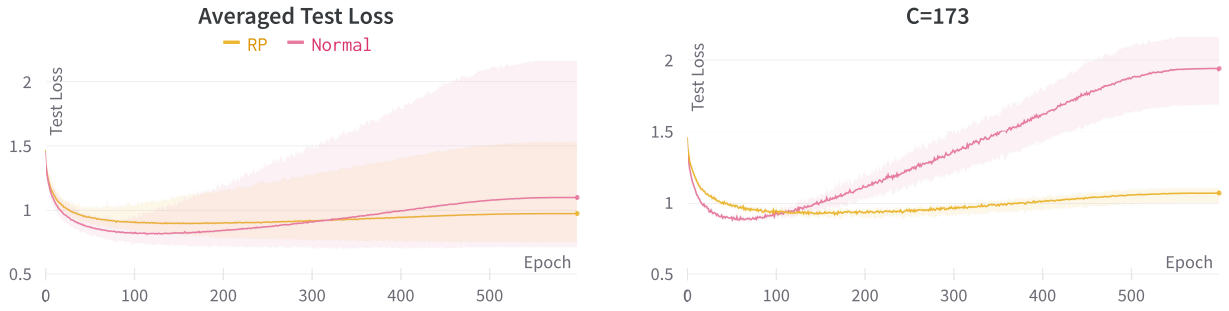

Figure S.6 . (Left) The average test loss curves are shown for $C = 16, 32, 64, 114, 173$ and five trials with different random seeds. The models used in this experiment were Normal and RP S-Mixers trained on CIFAR-10 with $L = 8$ for 600 epochs. The initial learning rate was set to 0.1. (Right) The test loss curve for $C = 173$ represents the worst case of overfitting. The shaded area in both figures represents the range between the maximum and minimum values.

Figure S.7 (right) illustrates that RP did not reach the relatively small training loss as the normal model. To address this, SGD was replaced with AdamW as the optimizer, with a reduced initial learning rate (lr $= 0.01$) due to the instability observed with lr $= 0.1$ in Figure S.8 . This resulted in reduced overfitting in the $C > S$ region, and RP performed exceptionally well compared to the normal model for $C = S = 64$. In Figure S.8 , neither the normal nor RP models exhibited a significant increase in test loss for $C = S$. However, while the normal model's test loss plateaued, the RP model continued to decrease its test loss, eventually surpassing the normal model in terms of test accuracy. This highlights the potential of RP to outperform the normal model with a choice of optimization such as AdamW.

D.8. Visualization of how random permutation breaks the structure

Observing the differences between RP-Mixer, MLP-Mixer, and SW-MLP from the perspective of weights. Figure S.9 shows three types of models where components can only be 0 or 1. In RP-Mixer, it is evident that the block structure of MLP-Mixer has been disrupted.

D.9. Computational Costs

Table S.3 shows the computational resource of SW-MLPs and RP-Mixers. In the setting for ImageNet, SW-MLP requires huge memory and runtime, whereas RP-Mixer needs 10^3 to 10^6 times less. Note that the spacial complexity of the

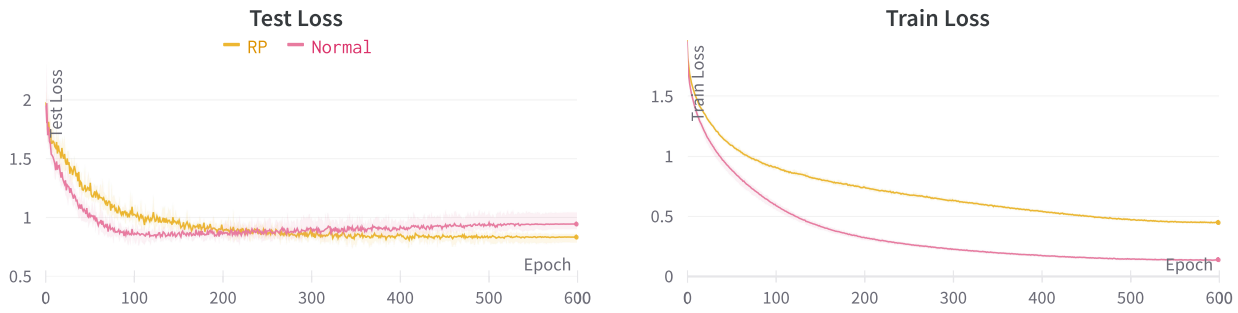

Figure S.7 . (Left) Average of test loss curves. (Right) Average train loss with $C = 173$. In both figures, the area shows max and min values. We trained normal and RP MLP-Mixer with $\gamma = 4$ with an initial learning rate of 0.1 and a mini-batch size of 256 for 600 epochs. The results are average of five trials. The shaded area in the figure represents the range of values between the maximum and minimum values.

SW-MLP (resp. the RP-Mixer) is $O(m^2) = O(S^2C^2)$ (resp. $O(C^2 + S^2)$) as $S, C \to \infty$. Therefore, the RP-Mixer is more memory-efficient and computationally undemanding than the SW-MLP.

Model	Memory(CIFAR/ImageNet)	FLOPs(CIFAR/ImageNet)	Runtime(CIFAR)
SW-MLP	3.22GB / 23.2TB	805M / 5.80T	$28.2(\pm 0.00)$ s/epoch
RP-Mixer	3.94MB / 26.3MB	12.5M / 4.06G	$6.41(\pm 0.01)$ s/epoch
MLP-Mixer	3.93MB / 26.3MB	12.5M / 4.06G	$6.08(\pm 0.26)$ s/epoch

Table S.3 . Comparison on memory requirements, FLOPs(floating point operations), and averaged runtime. For the three models, we set $S = 256, C = 588, L = 8, \gamma = 4$ for ImageNet and $S = 64, C = 48, L = 3, \gamma = 2$ for CIFAR.

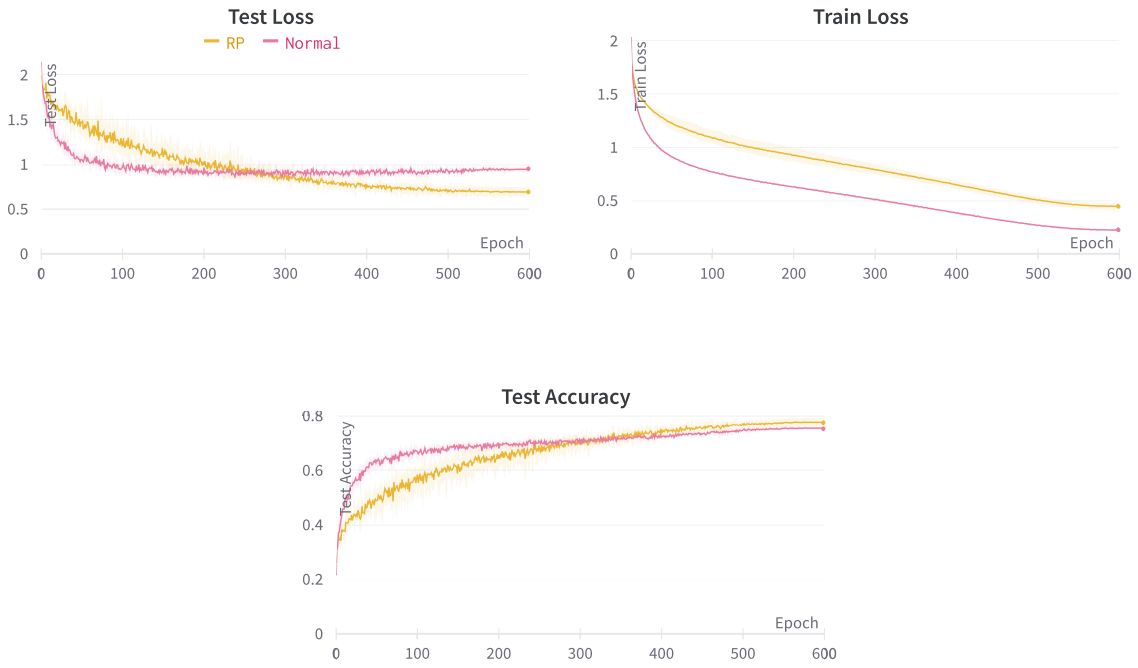

Figure S.8 . Results of training with AdamW and auto-augmentation with $S = C = 64$. The RP MLP-Mixer exceeded the results of the normal one in test loss and test accuracy. (Left) Average of test loss curves. (Right) Average train loss. (Lower) Test Accuracy curves. We set the initial learning rate to be 0.01 with a mini-batch size of 256 and 600 epochs. In all figures, the results are average of five trials and the area shows the range between the maximum and minimum values.

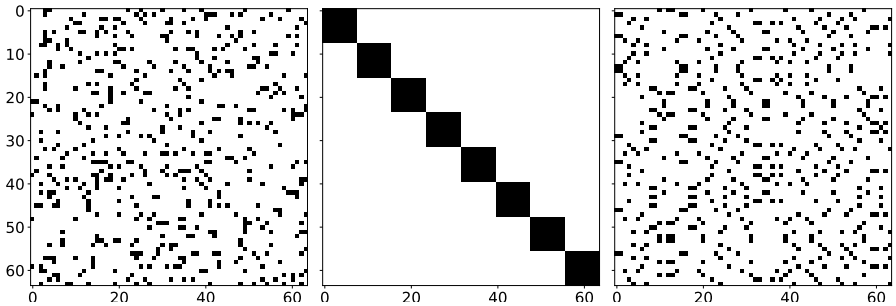

Figure S.9 . Samples of weight matrices at initialization. (Left) $M \odot A$, (center)$I_C \otimes W$, (right)$J_1(I_C \otimes W)J_2$, $C = S = 8, m = CS, p = 1/S$, and J_1 and J_2 are independent random permutation matrices. Each sample is a realization of one trial.